



F1

SCUDERIA FERRARI PERFORMANCE

Asifa Iqbal, Jocelynn Lopez, Alexa Lunn, & Daniel Parent

CONTEXT

Formual 1 is the highest class of international racing for single- seater, open-wheel cars. The “formula” in the name refers to the set of rules that all competing must adhere to.

Scuderia Ferrari is the longest-running and one of the most successful teams in F1 History. The current drivers for the 2025 season is the “Prince of Monaco”, Charles Leclerc and 7x World Champion, Lewis Hamilton.



2025 SEASON

The Ferrari SF-25 race car has been consistently underperforming in the 2025 Formula 1 season, demonstrating a significant lack of stability and a narrow operating window.

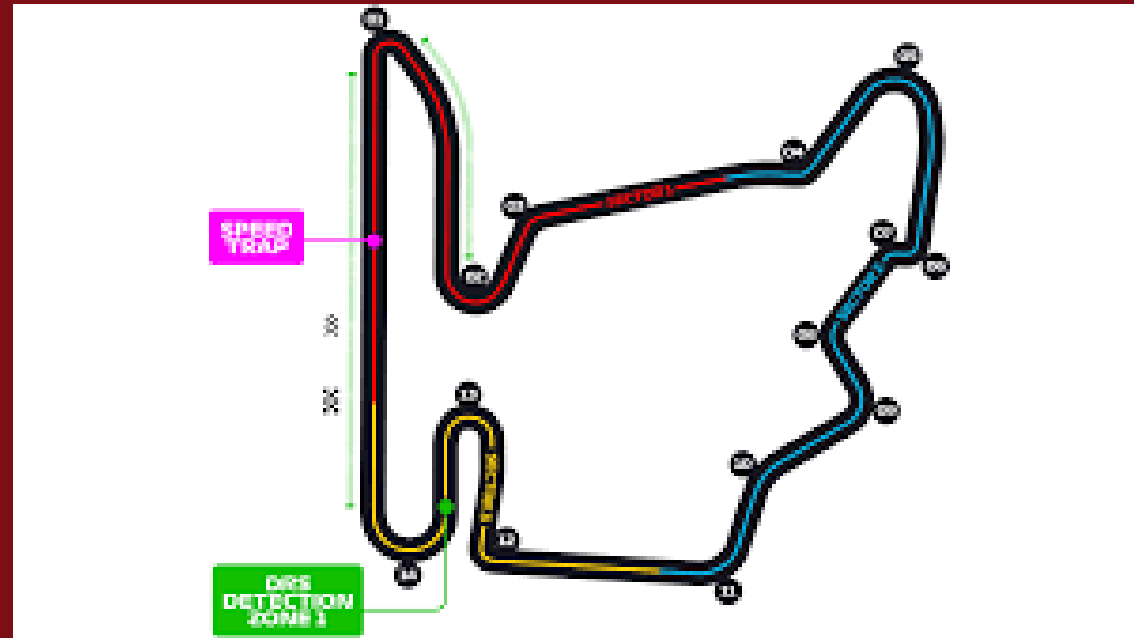
The car is highly sensitive to track conditions, ride height, and small setup changes, resulting in inconsistent performance between qualifying and race day and across different circuit types.

If Ferrari is able to pinpoint exactly what is causing these inconsistencies and resolve them to optimize the car for success on each circuit type, they can perform better when it matters which benefits everyone on the team at all levels.

how ferrari feels after aura farming the whole pre season and then qualifying behind albon in a williams:

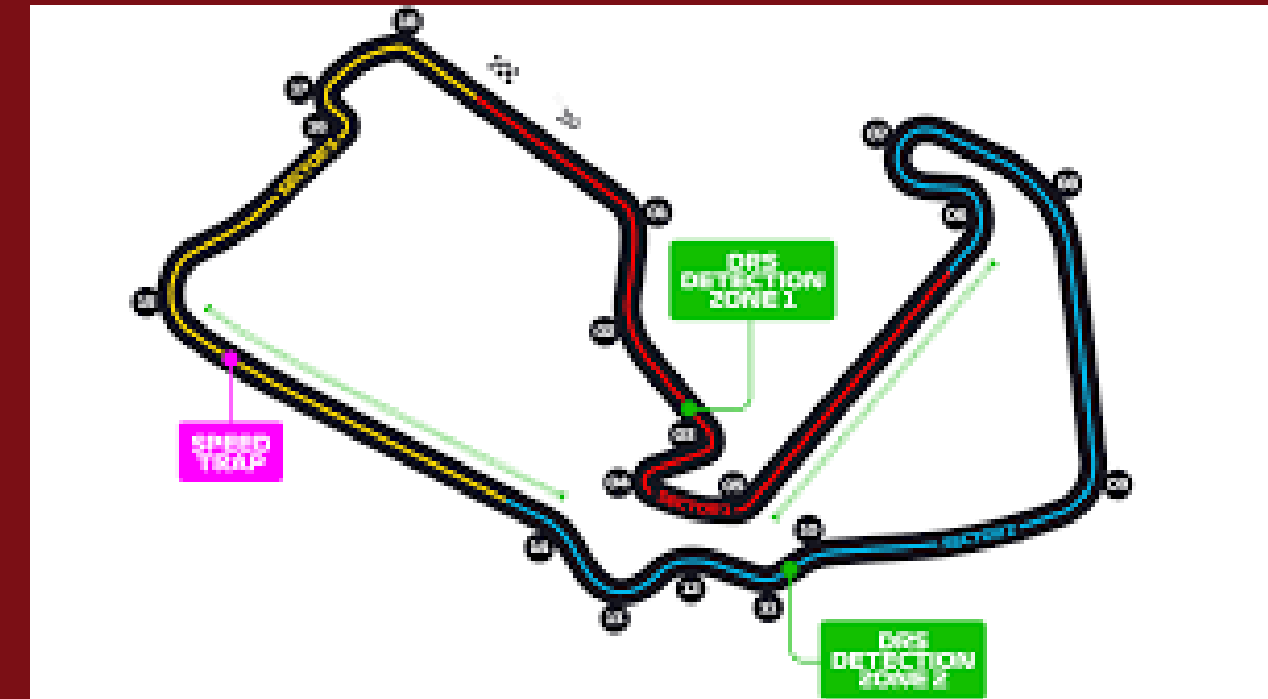


TRACKS THAT HURT THIS SEASON

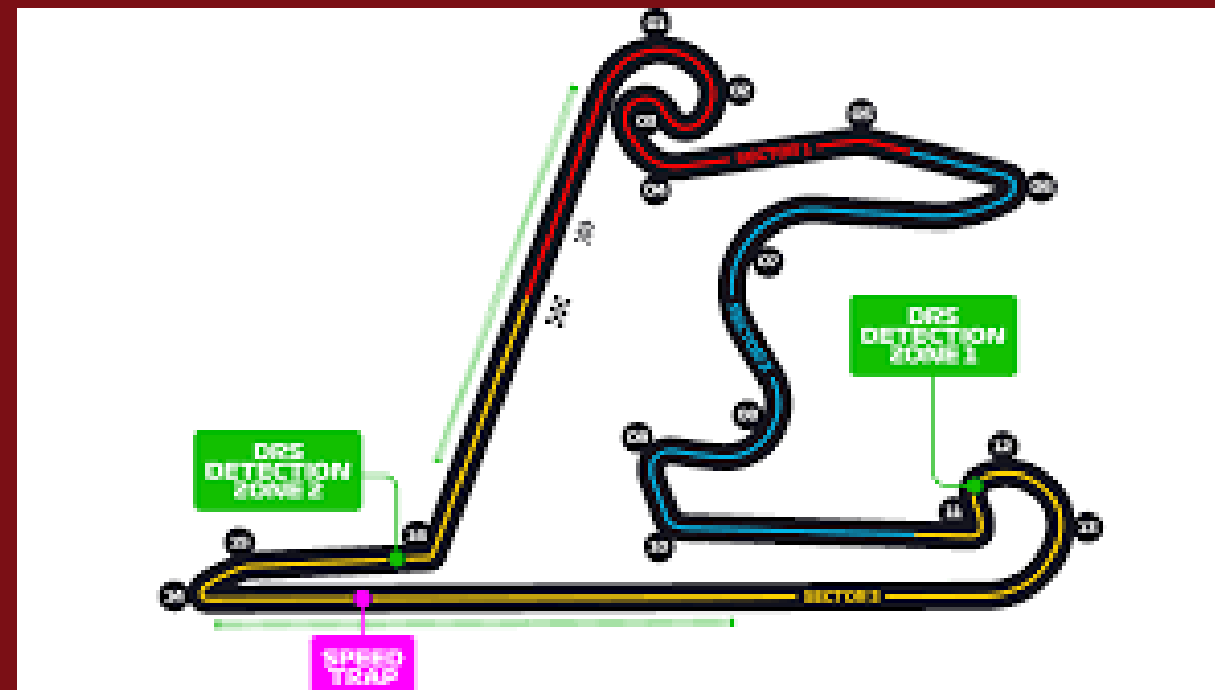


Hungarian Grand Prix
(Hungaroring)

Chinese Grand Prix
(Shanghai International Circuit)

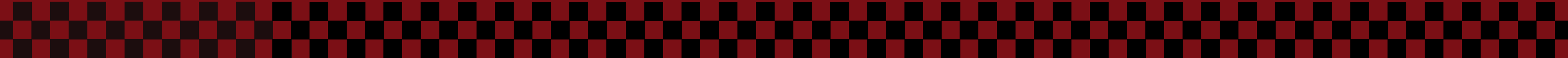


British Grand Prix
(Silverstone)





WHY CLEAN RELATIONAL DATA MATTERS

- **Accurate Modeling:** Dirty data pollutes advanced Computational Fluid Dynamics and simulator models, leading to a physical car (SF-25) that is inherently unstable and unpredictable
 - **Root Cause Analysis:** Clean, structurally sound relational data is essential to reliably pinpoint exact correlations between high-sensitivity variables (e.g., ride height, downforce, setup changes).
 - **Consistent Performance:** Inconsistent data causes the car to show inconsistent performance between sessions and circuits; clean data provides the confidence needed to make informed, aggressive setup adjustments for stability.
- 

ERD

Focused Entities for Our Problem

- We centered the design on **Driver**, **Race**, **Track**, and **Race Result** to capture the key factors behind Ferrari's inconsistent performance.

Supporting Insight Tables

- **Weather**, **Driver Track Experience**, **Bumpy Track Penalty**, and **Sensitivity Hotspot** help explain how conditions, rough tracks, and driver familiarity affect stability.

ERD Purpose

- The ERD connects all racing data into one system, making it easier to pinpoint what causes the SF-25's unpredictable behavior across circuits.

Key Relationship # 1

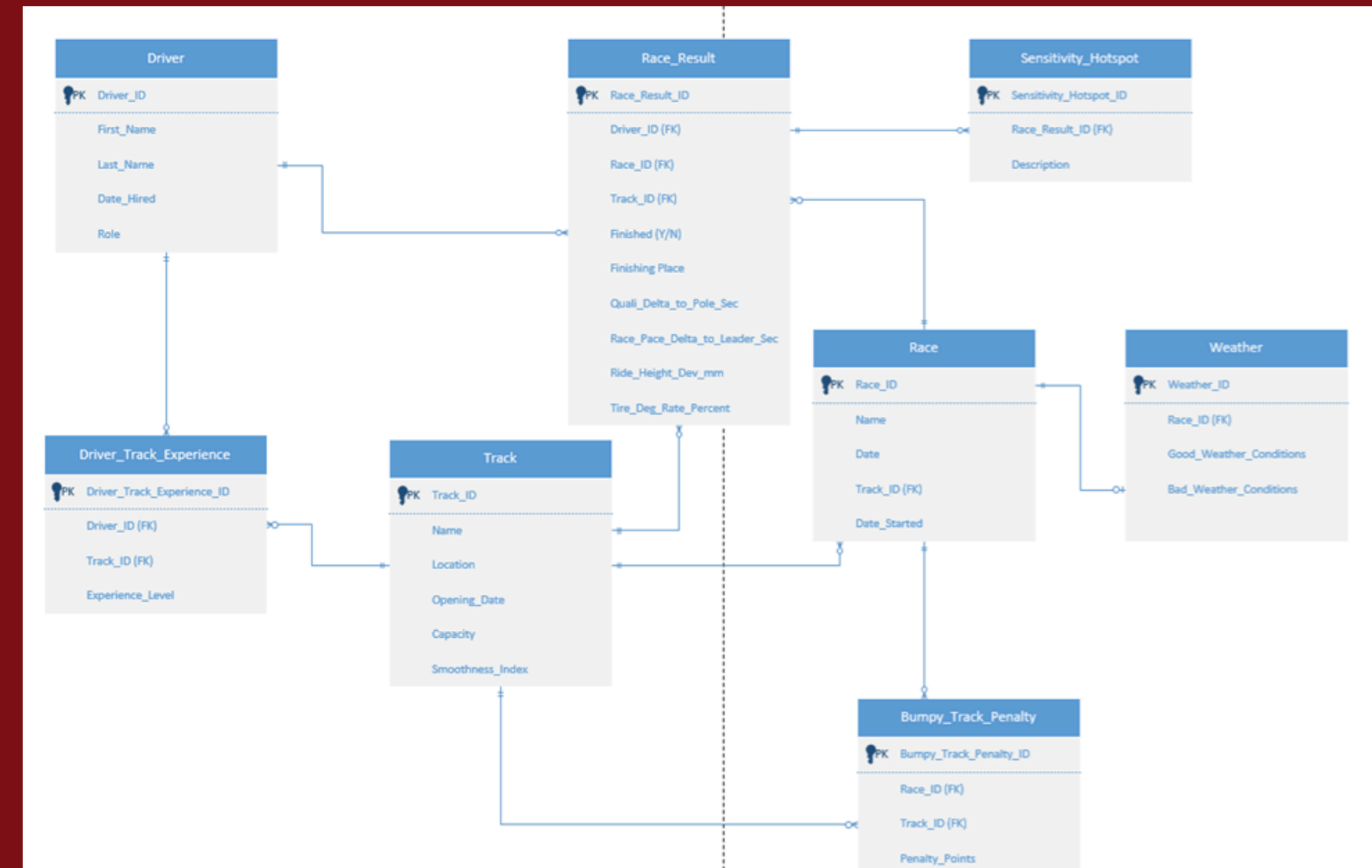
- Each **Race** occurs on one **Track**, linking track smoothness and conditions directly to performance results.

Key Relationship # 2

- A **Driver** has many **Race Results**, and each result can produce **Sensitivity Hotspots** that highlights where the car becomes unstable.

Key Relationship # 3

- **Tracks** generate multiple **Bumpy Track Penalty** records, showing how rough or uneven circuits contribute to the SF-25's instability during races.



THE “BUMPY TRACK PENALTY”

- Insight: The SF-25's performance gap to rivals significantly worsens on rough/bumpy circuits
 - The Bumpy_Penalty_Sec metric confirms the car's sensitivity to surface imperfections is a primary source of underperformance.
- Root Cause: The mechanical setup (suspension/dampers) fails to maintain a stable aerodynamic platform when the track surface is uneven.
- Actionable Step:
 - Focus: Mechanical & Suspension Kinematics.
 - Goal: Re-design suspension components to achieve a wider operating window that isolates the aero platform from track surface changes.

SETUP SENSITIVITY HOTSPOTS

- Insight: A strong correlation exists between high Ride_Height_Dev_mm and catastrophic Tire_Deg_Rate_Percent.
 - The Sensitivity_Metric highlights races where the car critically fell out of its ideal operating window.
- Root Cause: When ride height deviates even slightly (due to bumps or aero load changes), the floor stalls or becomes inefficient, causing high slip angles and rapidly destroying the tires.
- Actionable Step:
 - Focus: Aerodynamics & Correlation.
 - Goal: Analyze data and specific measurements from the highest-ranked 'Hotspot' races to pinpoint the exact corner/moment the ride height instability was triggered. This data is key for CFD/wind tunnel testing.

WALK-THROUGH OF SQL

```
WITH High_Sensitivity_Races AS (  
    -- Select races where both ride height deviation and tire degradation were above a performance threshold (e.g., > 1.5mm dev and > 2.0% deg)  
    SELECT  
        RR.Driver_ID,  
        RR.Race_ID,  
        RR.Ride_Height_Dev_mm,  
        RR.Tire_Deg_Rate_Percent,  
        (RR.Ride_Height_Dev_mm * RR.Tire_Deg_Rate_Percent) AS Sensitivity_Metric -- Custom metric to quantify severity  
    FROM  
        Race_Result RR  
    WHERE  
        RR.Ride_Height_Dev_mm > 1.5  
        AND RR.Tire_Deg_Rate_Percent > 2.00  
        AND RR.Finished = 'Y'  
    )  
SELECT
```

100 %

Results Messages

	Race_Name	Last_Name	Ride_Height_Dev_mm	Tire_Deg_Rate_Percent	Sensitivity_Metric	Sensitivity_Rank
1	Belgian Grand Prix	Fluxa	2.8	5.20	14.560	1
2	Austrian Grand Prix	Ilott	2.6	4.80	12.480	2
3	Dutch Grand Prix	Baldisserri	2.3	4.60	10.580	3
4	Emilia Romagna Grand Prix	Clear	2.1	4.20	8.820	4
5	United States Grand Prix	Leclerc	2.5	3.50	8.750	5
6	Las Vegas Grand Prix	Smith	2.1	4.10	8.610	6
7	Mexico City Grand Prix	Montoya	2.0	3.90	7.800	7
8	Chinese Grand Prix	Armstrong	1.8	3.60	6.480	8
9	Canadian Grand Prix	Crawford	1.7	3.30	5.610	9



DRIVER CONSISTENCY ANALYSIS

Insight: The Quali_Consistency_STDEV is high for both drivers, indicating the problem is with the car's intrinsic unpredictability, not the individual driver.

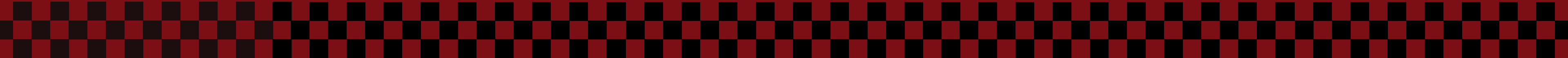
The car's optimal setup point is too narrow and moves drastically with minor environmental changes (ambient/track temperature).

Root Cause: The fundamental aerodynamic or mechanical design lacks robustness, making the car hard to setup consistently from one venue to the next.

Actionable Step:

Focus: Design Robustness.

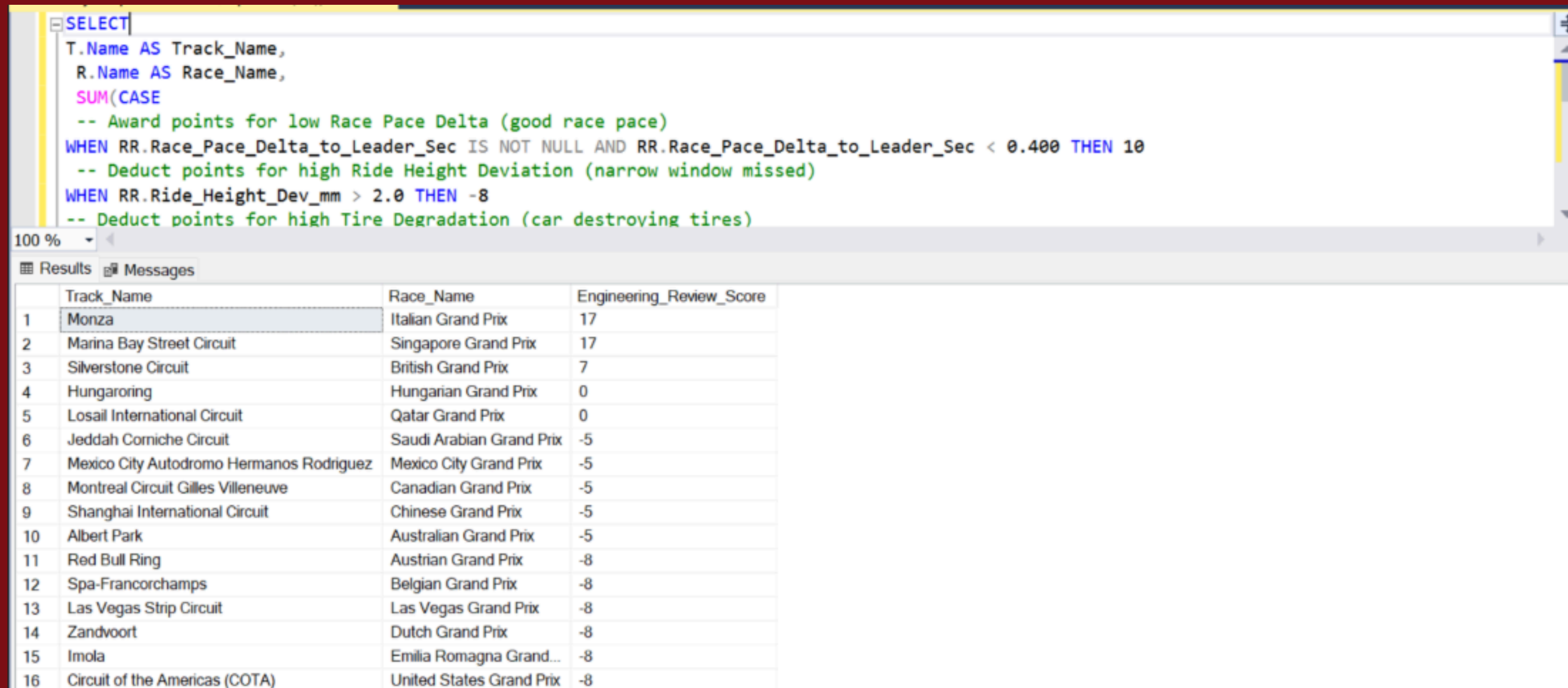
Goal: Invest in design revisions that lead to a flatter performance curve (wider plateau) for key setup parameters, minimizing the performance drop-off when the car is not perfectly optimized.



OVERALL PERFORMANCE SCORECARD

- Insight: The Engineering_Review_Score provides a single, weighted metric for comparing weekend success.
 - Negative scores are often driven by the ride height and tire degradation penalties (confirming the cause of the poor result).
- Root Cause: Confirms that race-losing weekends are predominantly driven by our specific technical weaknesses (ride height, tire wear) rather than a general lack of overall pace.
- Actionable Step:
 - Focus: Replication & Prevention.
 - Goal: Immediately replicate the setup parameters from high-scoring race weekends to build a baseline for the start of the wide

WALK-THROUGH OF SQL



The screenshot shows a SQL query editor with a query that calculates an 'Engineering_Review_Score' based on track and race names. The query uses a CASE statement to award points for low race pace delta and deduct points for high ride height deviation and high tire degradation. Below the query editor, the 'Results' tab is active, displaying a table with 16 rows of data. The table has four columns: an index, Track_Name, Race_Name, and Engineering_Review_Score.

```
SELECT
  T.Name AS Track_Name,
  R.Name AS Race_Name,
  SUM(CASE
    -- Award points for low Race Pace Delta (good race pace)
    WHEN RR.Race_Pace_Delta_to_Leader_Sec IS NOT NULL AND RR.Race_Pace_Delta_to_Leader_Sec < 0.400 THEN 10
    -- Deduct points for high Ride Height Deviation (narrow window missed)
    WHEN RR.Ride_Height_Dev_mm > 2.0 THEN -8
    -- Deduct points for high Tire Degradation (car destroying tires)
  ) AS Engineering_Review_Score
```

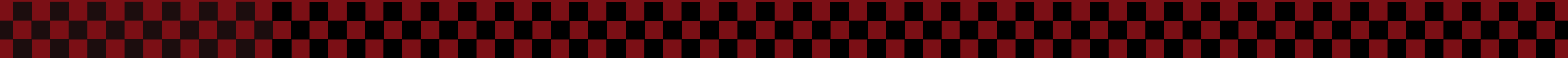
	Track_Name	Race_Name	Engineering_Review_Score
1	Monza	Italian Grand Prix	17
2	Marina Bay Street Circuit	Singapore Grand Prix	17
3	Silverstone Circuit	British Grand Prix	7
4	Hungaroring	Hungarian Grand Prix	0
5	Losail International Circuit	Qatar Grand Prix	0
6	Jeddah Corniche Circuit	Saudi Arabian Grand Prix	-5
7	Mexico City Autodromo Hermanos Rodriguez	Mexico City Grand Prix	-5
8	Montreal Circuit Gilles Villeneuve	Canadian Grand Prix	-5
9	Shanghai International Circuit	Chinese Grand Prix	-5
10	Albert Park	Australian Grand Prix	-5
11	Red Bull Ring	Austrian Grand Prix	-8
12	Spa-Francorchamps	Belgian Grand Prix	-8
13	Las Vegas Strip Circuit	Las Vegas Grand Prix	-8
14	Zandvoort	Dutch Grand Prix	-8
15	Imola	Emilia Romagna Grand Prix	-8
16	Circuit of the Americas (COTA)	United States Grand Prix	-8

STRUCTURED DATA VIEW

- Creating a structured view of racing data for an AI model to then analyze provides greater insight into the causes of performance inconsistencies.
 - It is clear that while some drivers are inherently more unpredictable and need greater training to improve consistency, the main issue is the car.
- Identified Causes:
 - Nearly all cases of poor performance were aligned with a narrow operating window score, meaning that only ideal conditions lead to desired results.
 - Unexpected tire degradation was another strong predictor of poor results, as the tires often did not behave as expected given the racing conditions.
- Actionable Steps:
 - Focus on widening the car's operating window, not just peak performance.
 - Improve prediction accuracy for tire degradation and act on new data.



STORED ANALYSIS PROCEDURE

- Creating and executing a procedure which analyzes race data and produces a dashboard of categorized statistics also helps identify inconsistencies.
 - There appears to be quite a bit of fluctuation in finishing position by race, with a standard deviation over 5 and a middling average position of about 9.8/20.
 - Identified Causes:
 - Car setups do not seem to be optimized for different race conditions and each unique track, as the average instability score is pretty high for most drivers.
 - The operating window seems to be pretty thin due to the high position volatility across races as well as from qualifying to the actual race for each driver.
 - Actionable Steps
 - Focus on widening the car's operating window and aim for consistency.
 - Devote more time to research and engineering to better prepare for each race.
- 

WALK-THROUGH OF SQL

EXEC sp_analyze_ferrari_performance										
100 %										
Results Messages										
	report_section	total_races_analyzed	drivers_analyzed	avg_finishing_position	avg_instability_score	total_dnfs	races_with_quali_race_inconsistency	races_with_tire_issues	dnf_percentage	
1	EXECUTIVE SUMMARY	14	12	9.470588	8.608537016068	1	0	0	5.56	
	report_section	track_type	races_count	avg_finish	avg_instability	avg_quali_race_gap	avg_tire_degradation	dnf_count	problem_severity	
1	TRACK TYPE ANALYSIS	Street_Circuit	18	9.470588	8.608537016068	0.458235	3.070588	1	LOW - Stable Performance	
	report_section	race_date	race_name	track_name	track_type	driver_name	target_finish_position	narrow_operating_window_score	quali_race_performance_gap	tire_deg_deviation_from_avg
										finishing
	report_section	smoothness_category	race_count	avg_finish	avg_instability	avg_tire_deg	finish_position_stddev			
1	TRACK SMOOTHNESS IMPACT	0-39: Bumpy	18	9.470588	8.608537016068	3.070588	4.98895839653459			
	report_section	driver_name	races	avg_finish	avg_instability	avg_quali_race_gap	dnfs	avg_tire_deg	consistency_rating	
1	DRIVER PERFORMANCE COMPARISON	Leclerc	4	4.000000	12.6331563540931	0.300000	0	2.100000	Very Consistent	
2	DRIVER PERFORMANCE COMPARISON	Baldisserrri	1	15.000000	12.355684733705	0.650000	0	4.600000	Very Inconsistent	
3	DRIVER PERFORMANCE COMPARISON	Fluxa	1	18.000000	12.1850314495801	0.900000	0	5.200000	Very Inconsistent	
4	DRIVER PERFORMANCE COMPARISON	Hamilton	4	5.000000	11.0348330630769	0.220000	1	1.966666	Very Consistent	
5	DRIVER PERFORMANCE COMPARISON	Montoya	1	14.000000	9.93538460737146	0.650000	0	3.900000	Very Inconsistent	
6	DRIVER PERFORMANCE COMPARISON	Alesi	1	8.000000	9.52987457146399	0.400000	0	2.400000	Very Inconsistent	
7	DRIVER PERFORMANCE COMPARISON	Ilott	1	16.000000	4.77728827066554	0.700000	0	4.800000	Very Inconsistent	
8	DRIVER PERFORMANCE COMPARISON	Clear	1	13.000000	3.475	0.550000	0	4.200000	Very Inconsistent	
	report_section	priority	recommendation	supporting_data						
1	RECOMMENDATIONS	1	Focus setup development on Street_Circuit circuits	Avg Instability: 8.61						
2	RECOMMENDATIONS	4	Expand operating window testing	High position volatility suggests narrow setup w...						



ACQUIRED SKILLS

- Core SQL
 - Build out a relational database
 - Defining all the tables, primary keys, and relationships
 - Advanced SQL
 - Conditional Aggregation to precisely calculate key metrics
 - Data Engineering skills to automate processes
 - Stored Procedures
 - Feature Views
 - Leveraged AI tools to solve a unique real-world problem
 - Gemini
 - ChatGPT
 - Claude
- 